PROBLEM10:

PROBLEM STATEMENT:

University Admission Decision Engine: Evaluate candidates using subject eligibility, aggregate marks, course-specific thresholds, and tie-breaking criteria; generate a ranked list of eligible applicants.

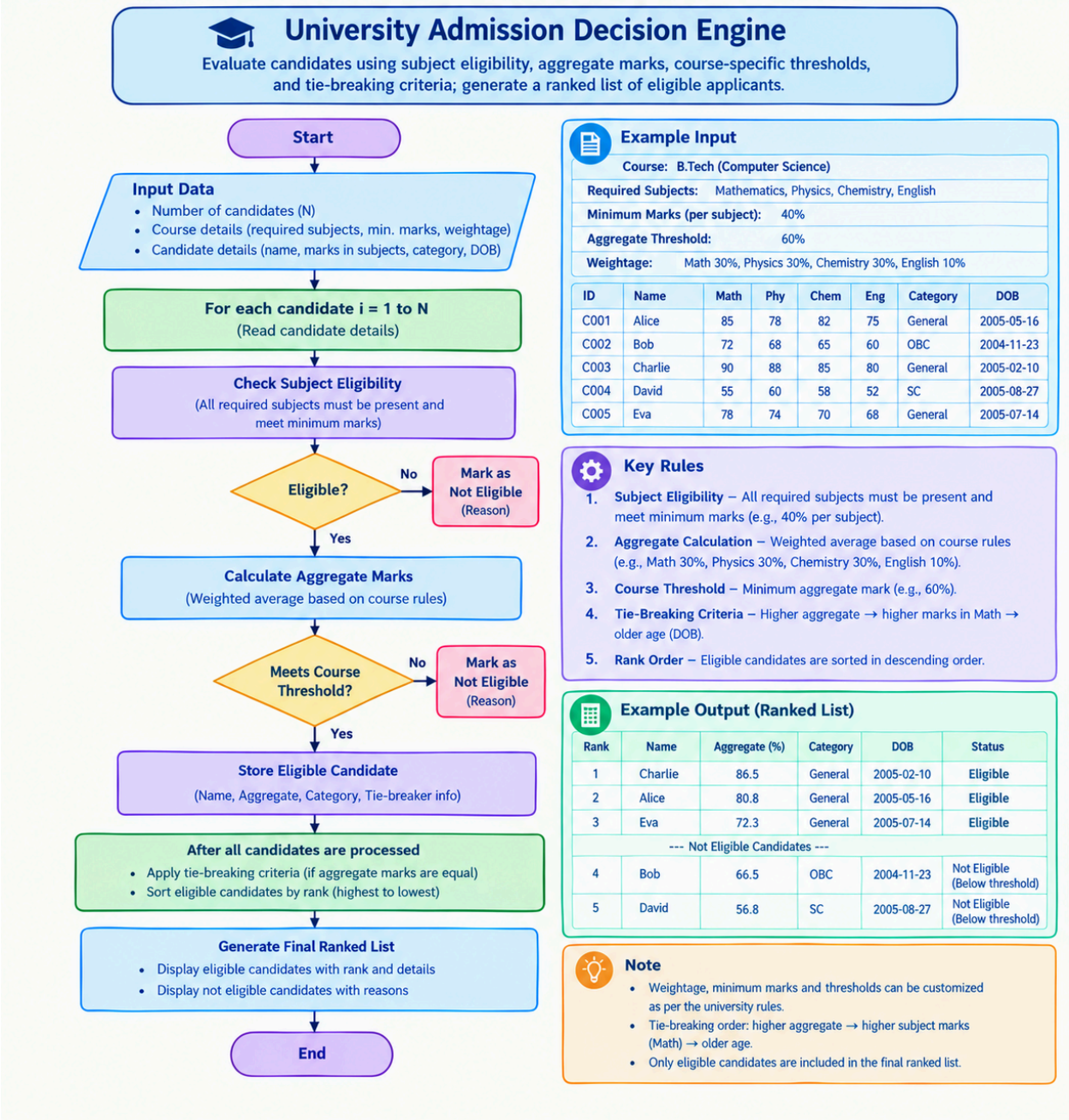
OBJECTIVE:

To develop a C program that evaluates university applicants based on subject eligibility, aggregate marks, course-specific cut-off marks, and tie-breaking criteria. The program identifies eligible candidates and generates a ranked merit list according to their performance.

ALGORITHM:

1. Start the program.
2. Input the number of applicants.
3. For each applicant, enter the candidate's name, subject marks, and course applied for.
4. Validate all marks to ensure they are between 0 and 100.
5. Calculate the aggregate percentage of each candidate.
6. Check subject-specific eligibility requirements for the selected course.
7. Compare the aggregate percentage with the course-specific cut-off.
8. Mark applicants who satisfy all eligibility requirements as eligible.
9. Rank eligible applicants by aggregate percentage in descending order.
10. If two candidates have the same aggregate, use the relevant subject mark as the tie-breaker.
11. Display the eligible applicants in ranked order, along with their marks and eligibility status.
12. Stop the program.

FLOWCHART:



PROGRAM:

The program below uses three sample courses with different eligibility rules. If two eligible candidates have the same aggregate, the candidate with the higher course-specific subject mark is ranked first.

```
#include <stdio.h>
#include <string.h>

#define MAX 50

struct Applicant {
    char name[30];
    float english, maths, science, aggregate;
    int eligible;
};

int main() {
    struct Applicant a[MAX], temp;
    int n, i, j, tieSubject;
    char course[20];
    float cutoff, subjectMin;

    printf("Enter course (Engineering/Science/Arts): ");
    scanf("%19s", course);

    if (strcmp(course, "Engineering") == 0) {
        cutoff = 70;
        subjectMin = 60;
        tieSubject = 1;  // Maths
    } else if (strcmp(course, "Science") == 0) {
        cutoff = 60;
        subjectMin = 50;
        tieSubject = 2;  // Science
    } else if (strcmp(course, "Arts") == 0) {
        cutoff = 50;
        subjectMin = 40;
        tieSubject = 0;  // English
    } else {
        printf("Invalid course!\n");
        return 0;
    }

    printf("Enter number of applicants (1-%d): ", MAX);
    scanf("%d", &n);
```

```
if (n < 1 || n > MAX) {
    printf("Invalid number of applicants!\n");
    return 0;
}

for (i = 0; i < n; i++) {
    printf("\nApplicant %d name: ", i + 1);
    scanf("%29s", a[i].name);

    printf("English marks: ");
    scanf("%f", &a[i].english);

    printf("Maths marks: ");
    scanf("%f", &a[i].maths);

    printf("Science marks: ");
    scanf("%f", &a[i].science);

    if (a[i].english < 0 || a[i].english > 100 ||
        a[i].maths < 0 || a[i].maths > 100 ||
        a[i].science < 0 || a[i].science > 100) {
        printf("Invalid marks! Enter values from 0 to 100.\n");
        return 0;
    }

    a[i].aggregate =
        (a[i].english + a[i].maths + a[i].science) / 3.0;

    float requiredMark;

    if (tieSubject == 0)
        requiredMark = a[i].english;
    else if (tieSubject == 1)
        requiredMark = a[i].maths;
    else
        requiredMark = a[i].science;

    a[i].eligible =
        (a[i].aggregate >= cutoff &&
         requiredMark >= subjectMin);
}

/* Sort eligible applicants before ineligible applicants.
```

```
   Rank by aggregate, then by course-specific subject mark. */
for (i = 0; i < n - 1; i++) {
    for (j = i + 1; j < n; j++) {
        float markI, markJ;

        if (tieSubject == 0) {
            markI = a[i].english;
            markJ = a[j].english;
        } else if (tieSubject == 1) {
            markI = a[i].maths;
            markJ = a[j].maths;
        } else {
            markI = a[i].science;
            markJ = a[j].science;
        }

        if (a[j].eligible > a[i].eligible ||
            (a[j].eligible == a[i].eligible &&
             a[j].aggregate > a[i].aggregate) ||
            (a[j].eligible == a[i].eligible &&
             a[j].aggregate == a[i].aggregate &&
             markJ > markI)) {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
        }
    }
}

printf("\n--- ADMISSION MERIT LIST: %s ---\n", course);
printf("Aggregate cutoff: %.2f%%\n", cutoff);
printf("Subject minimum: %.2f\n\n", subjectMin);

printf("Rank  Name       Aggregate   Status\n");
printf("---------------------------------------");

int rank = 0;

for (i = 0; i < n; i++) {
    if (a[i].eligible) {
        rank++;
        printf("\n%-5d %-10s %-11.2f Eligible",
               rank, a[i].name, a[i].aggregate);
    }
```

```
    }

    if (rank == 0)
        printf("\nNo eligible applicants.");

    printf("\n\nTotal eligible applicants: %d\n", rank);

    return 0;
}
```

PSEUDOCODE:

```
BEGIN

INPUT course

IF course = "Engineering"
    cutoff = 70
    subject minimum = 60
    tie-breaker = Maths
ELSE IF course = "Science"
    cutoff = 60
    subject minimum = 50
    tie-breaker = Science
ELSE IF course = "Arts"
    cutoff = 50
    subject minimum = 40
    tie-breaker = English
ELSE
    DISPLAY "Invalid course"
    STOP
END IF

INPUT number of applicants

FOR each applicant
    INPUT name
    INPUT English marks
    INPUT Maths marks
    INPUT Science marks

    VALIDATE all marks are between 0 and 100
```

```
    aggregate = (English + Maths + Science) / 3

    CHECK course-specific subject requirement

    IF aggregate >= cutoff AND subject requirement is met
        eligible = TRUE
    ELSE
        eligible = FALSE
    END IF
END FOR

SORT applicants:
    Eligible applicants first
    Higher aggregate first
    Higher tie-breaker subject mark first

DISPLAY eligible applicants in ranked order
DISPLAY total number of eligible applicants

END
```

OUTPUT:

Example: Engineering admission

Enter course (Engineering/Science/Arts): Engineering
Enter number of applicants (1-50): 4

Applicant 1 name: Ravi
English marks: 75
Maths marks: 85
Science marks: 80

Applicant 2 name: Sita
English marks: 80
Maths marks: 90
Science marks: 85

Applicant 3 name: Arun
English marks: 65
Maths marks: 55
Science marks: 70

Applicant 4 name: Priya
English marks: 75
Maths marks: 85
Science marks: 80

--- ADMISSION MERIT LIST: Engineering ---
Aggregate cutoff: 70.00%
Subject minimum: 60.00

Rank Name Aggregate Status
---
1 Sita 85.00 Eligible
2 Ravi 80.00 Eligible
3 Priya 80.00 Eligible

Total eligible applicants: 3